stationary at this minimum is just that ϕ^Λ should satisfy Eq. (27.4.11). We see then that in the absence of Fayet–Iliopoulos D-terms, the question of whether supersymmetry is unbroken in gauge theories is entirely a matter of whether the superpotential allows solutions of Eq. (27.4.10). The same result applies even in non-renormalizable theories.[3]

Now let us assume that there is a set of values ϕ_{n0} at which $V(\phi_0) = 0$, so that supersymmetry is unbroken. The spin 0 degrees of freedom are described by a shifted field

$$\varphi_n = \phi_n - \phi_{n0}\,. \tag{27.4.13}$$

There is then a cross-term between φ and the gauge fields, arising from the first term in Eq. (27.4.1):

$$2\sum_{nA} \mathrm{Im}\left(\partial_\mu \varphi_n\, (t_A \phi_0)^*_n\right) V^\mu_A\,.$$

As shown in Section 21.1, it is always possible to eliminate this term by adopting a 'unitarity gauge,' in which ϕ_n satisfies a constraint that makes this term vanish:

$$\sum_n \mathrm{Im}\left(\phi_n\, (t_A \phi_0)^*_n\right) = 0\,. \tag{27.4.14}$$

This will have the effect of eliminating the Goldstone bosons associated with broken gauge symmetries.

We shall now work out the masses of the particles of spin 0, spin 1/2, and spin 1 that arise in this theory if supersymmetry is unbroken, taking into account the possibility that the gauge symmetries may be spontaneously broken.

Spin 0

Because $\partial f(\phi)/\partial \phi_n$ and $\xi_A + \sum_{nm} \phi^*_n (t_A)_{nm}\, \phi_m$ must both vanish at $\phi_n = \phi_{n0}$, the terms in $V(\phi)$ of second order in $\varphi_n \equiv \phi_n - \phi_{n0}$ and/or φ^*_n are of the form

$$\begin{aligned} V_{\text{quad}}(\phi) = {} & \sum_{nm} (\mathcal{M}^* \mathcal{M})_{nm} \varphi^*_n \varphi_m + \sum_{Anm} \left(t_A \phi_0\right)_n \left(t_A \phi_0\right)^*_m \varphi^*_n \varphi_m \\ & + \frac{1}{2} \sum_{Anm} \left(t_A \phi_0\right)^*_n \left(t_A \phi_0\right)^*_m \varphi_n \varphi_m + \frac{1}{2} \sum_{Anm} \left(t_A \phi_0\right)_n \left(t_A \phi_0\right)_m \varphi^*_n \varphi^*_m\,, \end{aligned} \tag{27.4.15}$$

where $\mathcal{M}$ is the complex symmetric matrix (26.4.11):

$$\mathcal{M}_{nm} \equiv \left(\frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m}\right)_{\phi=\phi_0}\,.$$